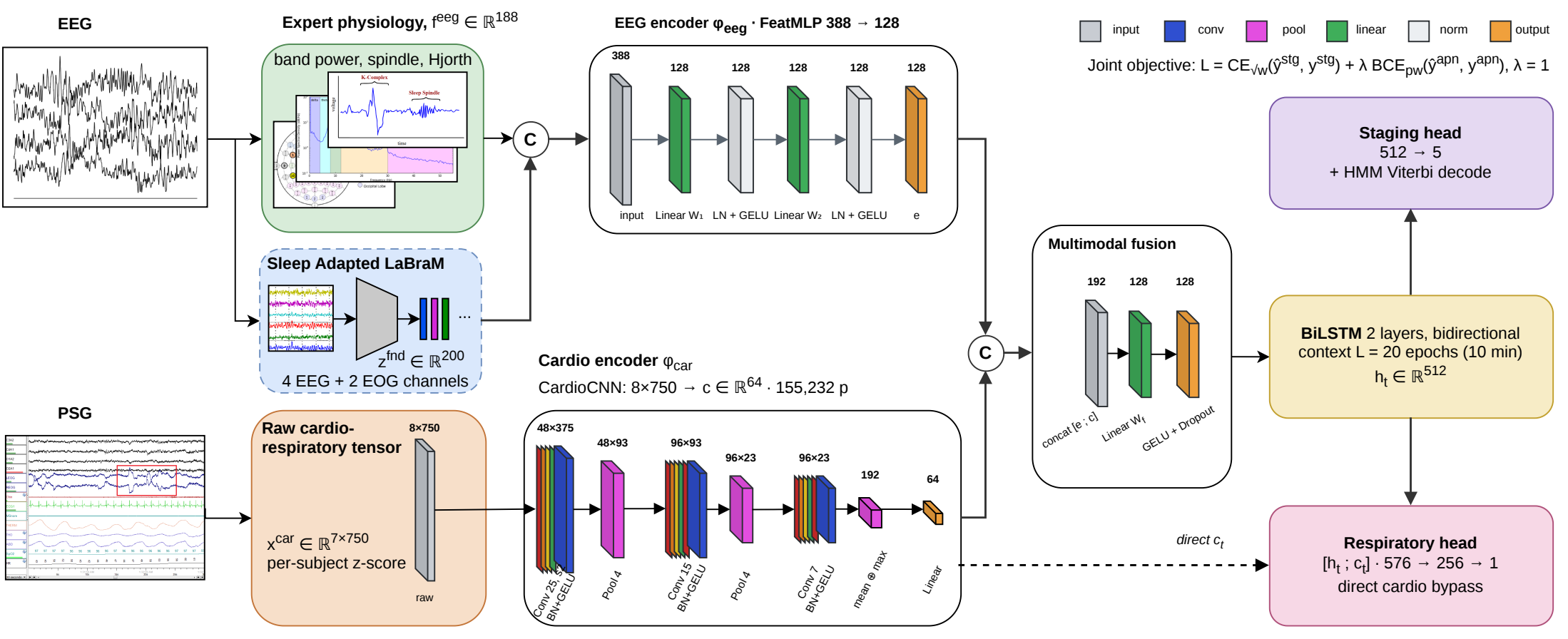

EEG
Expert physiology, f^eeg ∈ ℝ^188
band power, spindle, Hjorth
K-Complex
Sleep Spindle
EEG encoder φ_eeg · FeatMLP 388 → 128
388
128
128
128
128
128
input
Linear W₁
LN + GELU
Linear W₂
LN + GELU
e
input
conv
pool
linear
norm
output
Joint objective: L = CE_√w(ŷ^stg, y^stg) + λ BCE_pw(ŷ^apn, y^apn), λ = 1
Staging head
512 → 5
+ HMM Viterbi decode
Sleep Adapted LaBraM
z^fnd ∈ ℝ^200
4 EEG + 2 EOG channels
Multimodal fusion
192
128
128
concat [e ; c]
Linear W_f
GELU + Dropout
BiLSTM 2 layers, bidirectional
context L = 20 epochs (10 min)
h_t ∈ ℝ^512
PSG
Raw cardio-respiratory tensor
8×750
x^car ∈ ℝ^7×750
per-subject z-score
raw
Cardio encoder φ_car
CardioCNN: 8×750 → c ∈ ℝ^64 · 155,232 p
48×375
48×93
96×93
96×23
96×23
192
64
Conv 25, s=5 BN+GELU
Pool 4
Conv 15 BN+GELU
Pool 4
Conv 7 BN+GELU
mean ⊕ max
Linear
direct c_t
Respiratory head
[h_t ; c_t] · 576 → 256 → 1
direct cardio bypass